Ejemplo

Resolver

$$y''' - 6y'' + 12y' - 8y = 0$$

Sujeta a las condiciones iniciales:

$$y(0) = 2$$
$$y'(0) = 1$$
$$y''(0) = 0$$

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$y = e^{rx} \quad y' = re^{rx} \quad y'' = r^2 e^{rx} \quad y''' = r^3 e^{rx}$

$r^3 e^{rx} - 6r^2 e^{rx} + 12r e^{rx} - 8e^{rx} = 0$

$e^{rx}(r^3 - 6r^2 + 12r - 8) = 0$

$r^3 - 6r^2 + 12r - 8 = 0$

$(r-2)^3 = 0 \qquad (r-2)(r-2)(r-2) = 0$

$r = 2$

$e^{2x} \quad x e^{2x} \quad x^2 e^{2x} \qquad a = 2$

$y_1 = e^{2x} \qquad y_2 = x e^{2x} \qquad y_3 = x^2 e^{2x}$

$$y = c_1 e^{2x} + c_2 x e^{2x} + c_3 x^2 e^{2x}$$ solución general

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$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix}$$

$y_1 = e^{2x} \qquad y_2 = x e^{2x} \qquad y_3 = x^2 e^{2x}$

$y_1' = 2e^{2x} \qquad y_2' = e^{2x} + 2xe^{2x} \qquad y_3' = 2xe^{2x} + 2x^2 e^{2x}$

$y_1'' = 4e^{2x} \qquad y_2'' = 4e^{2x} + 4xe^{2x} \qquad y_3'' = 2e^{2x} + 8xe^{2x} + 4x^2 e^{2x}$

$$W = \begin{vmatrix} e^{2x} & xe^{2x} & x^2 e^{2x} \\ 2e^{2x} & e^{2x} + 2xe^{2x} & 2xe^{2x} + 2x^2 e^{2x} \\ 4e^{2x} & 4e^{2x} + 4xe^{2x} & 2e^{2x} + 8xe^{2x} + 4x^2 e^{2x} \end{vmatrix}$$

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$e^{2x} \cdot e^{2x} \cdot e^{2x} = e^{6x}$

$$W = e^{6x} \cdot \begin{vmatrix} 1 & x & x^2 \\ 2 & 1+2x & 2x+2x^2 \\ 4 & 4+4x & 2+8x+4x^2 \end{vmatrix}$$

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$x = 0$

$$W(0) = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 4 & 2 \end{vmatrix} = 1 \cdot 1 \cdot 2 = 2 \qquad W \neq 0$$

$$y = c_1 e^{2x} + c_2 x e^{2x} + c_3 x^2 e^{2x}$$

$y(0) = 2$
$y'(0) = 1$
$y''(0) = 0$

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$y(0) = \boxed{c_1 = 2}$

$y' = 2 \cdot c_1 e^{2x} + c_2(e^{2x} + 2xe^{2x}) + c_3(2xe^{2x} + 2x^2 e^{2x})$

$y'(0) = 2 \cdot c_1 + c_2 = 1$

$2 \cdot 2 + c_2 = 1 \longrightarrow \boxed{c_2 = -3}$

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$y'' = 4c_1 e^{2x} + c_2(4e^{2x} + 4xe^{2x}) + c_3(2e^{2x} + 8xe^{2x} + 4x^2 e^{2x})$

$y''(0) = 4 \cdot c_1 + 4 \cdot c_2 + 2 \cdot c_3 = 0$

$4 \cdot 2 + 4 \cdot (-3) + 2 \cdot c_3 = 0$

$8 - 12 + 2 \cdot c_3 = 0$

$-4 + 2c_3 = 0$

$2c_3 = 4 \longrightarrow \boxed{c_3 = 2}$

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$$y = c_1 e^{2x} + c_2 x e^{2x} + c_3 x^2 e^{2x}$$

$$\boxed{y = 2e^{2x} - 3xe^{2x} + 2x^2 e^{2x}}$$

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